

Multi-UPF Traffic Steering for 5G Network Slices in Stadium and Mass Event Deployments

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Abstract—Mass-event venues such as stadiums impose extreme and heterogeneous demands on 5G network infrastructure: enhanced Mobile Broadband (eMBB) sessions for high-definition fan engagement, Ultra-Reliable Low-Latency Communication (URLLC) for latency-sensitive augmented- and virtual-reality overlays, and massive Machine-Type Communication (mMTC) for pervasive crowd sensing must coexist at peak load. A single centralised User Plane Function (UPF) quickly becomes a bottleneck in such scenarios, motivating a multi-UPF, multi-access edge computing (MEC) architecture in which session-to-UPF assignments are made dynamically. This paper proposes and evaluates an Evolutionary Game Theory (EGT) replicator-dynamics controller within the 5G Session Management Function (SMF) to steer traffic across an edge UPF and a central-cloud UPF in a slice-aware, load-adaptive manner. The controller is validated on an OpenAirInterface (OAI) emulation testbed supporting 1,000 concurrent PDU sessions partitioned into an 800-UE eMBB slice and a 200-UE URLLC slice, achieving 100% PDU session establishment success at scale via a high-performance Go-based user-plane multiplexing engine. Results show that the EGT controller achieves zero eMBB Packet Delay Budget (PDB) violations in standard and overload scenarios, and converges to a stable equilibrium in under 100 iterations across all evaluated conditions. During a $5.5\times$ halftime demand surge, model-projected aggregate throughput approaches 18 Gbps, saturating edge-node processing capacity and revealing a hard infrastructure boundary for the 10 ms URLLC PDB that is independent of steering policy. Future work will investigate a Bidirectional LSTM (BLSTM) predictor as a proactive reconfiguration trigger, with the goal of reducing reactive signalling overhead.

Index Terms—5G network slicing, User Plane Function (UPF) selection, multi-access edge computing (MEC), quality of service (QoS), radio access network (RAN), evolutionary game theory (EGT), OpenAirInterface

ABBREVIATIONS

Abbreviation	Full Form
5G	Fifth Generation
eMBB	Enhanced Mobile Broadband
EGT	Evolutionary Game Theory
MEC	Multi-Access Edge Computing
OAI	OpenAirInterface
PDB	Packet Delay Budget
QoS	Quality of Service
RAN	Radio Access Network
SMF	Session Management Function
UE	User Equipment
UPF	User Plane Function
URLLC	Ultra-Reliable Low-Latency Communication

I. INTRODUCTION

Stadiums and large-scale public events are increasingly becoming testbeds for next-generation connectivity. As 5G NR deployments mature, the distinct service classes they support open up possibilities that were simply not feasible with previous network generations: eMBB for high-throughput fan engagement, URLLC for mission-critical operations such as drone coordination, and mMTC for dense IoT sensing. Delivering immersive AR/VR experiences to tens of thousands of spectators simultaneously, while also supporting real-time safety systems and pervasive crowd monitoring, places extraordinary demands on network infrastructure. Latencies must remain below 20 ms for time-sensitive applications, throughput must scale to gigabit levels, and reliability cannot drop below 99.999% for safety-critical slices. Meeting all of these requirements in a single venue, at peak load, is a genuinely hard problem.

A key part of the solution lies at the edge. By deploying MEC nodes close to the RAN, operators can offload compute-intensive tasks to local servers, including 360° video processing, AI-based crowd analytics, and real-time resource orchestration. Moving this processing away from the core cuts round-trip latency significantly and improves the Quality of Experience (QoE) for end users. The growing availability of localised UPFs, driven in part by the broader expansion of distributed data centres, makes this architecture increasingly practical. However, relying on a single centralised UPF in a high-density scenario quickly becomes a bottleneck: band-

width gets congested, propagation delays creep up, and the QoS guarantees associated with each network slice begin to erode. A multi-UPF deployment addresses this by steering traffic to whichever edge node is best placed to serve a given UE, provided the selection logic is sound. Getting that logic right requires algorithms that can jointly account for load, latency, per-slice QoS requirements, and fairness across UEs, so that neither eMBB fans nor URLLC safety services are starved of resources. This paper proposes and evaluates such an algorithm, with the aim of enabling reliable 5G edge service delivery across all slice types in mass event environments.

II. RELATED WORK

The transition from hardware-centric legacy infrastructure to the 5G System (5GS) is fundamentally shaped by the adoption of a Service-Based Architecture (SBA). A cornerstone of this shift is Control and User Plane Separation (CUPS), which decouples session management logic from data-plane routing, enabling the UPF to be instantiated anywhere in the topology—from centralised data centres to the network edge. This architectural flexibility transforms UPF selection from a static assignment into a dynamic, slice-aware optimisation problem, and provides the architectural motivation for the work surveyed in this section.

The body of work relevant to this paper spans three interconnected areas: 5G network slicing and resource management, UPF placement and selection in mobile core networks, and MEC deployment in high-density environments.

Network slicing has been extensively studied since its introduction as a core concept in 3GPP Release 15. Early work focused on the radio access side, with proposals for slice-aware scheduling and resource partitioning at the base station level [?]. More recent contributions have addressed end-to-end slice management, including control-plane signalling, isolation guarantees, and dynamic admission control across eMBB, URLLC, and mMTC traffic classes [?]. These slices impose distinct KPI targets: eMBB demands peak data rates of 20 Gbps downlink and 10 Gbps uplink; URLLC mandates end-to-end latency below 1 ms with 99.999% availability; and mMTC targets connection densities of up to 10^6 devices/km² [?]. While these works establish the slicing framework, they largely assume a fixed core network topology and do not consider UPF placement as a variable in the optimisation. More recently, distributed QoS-aware slicing algorithms have been extended beyond 5G; [?] proposes an ADMM-based framework combining graph neural networks with distributed optimisation to achieve provably convergent resource slicing under dynamic and unreliable network conditions.

UPF placement and traffic steering have received growing attention as operators move toward distributed core architectures. Several studies have framed UPF placement as a facility location problem, optimising for latency or load under capacity constraints [?]. Others have proposed dynamic UPF migration triggered by mobility patterns or traffic shifts [?]. However, most of these approaches target general LTE/5G

deployments and do not account for the extreme density and heterogeneous QoS profiles characteristic of mass event scenarios, where eMBB, URLLC, and mMTC traffic coexist and compete for the same edge infrastructure. A promising EGT-based approach for dynamic UPF selection, where UEs adapt their choice to minimise end-to-end delay using replicator dynamics converging to a stable equilibrium, is proposed in [?]. Complementing this, [?] proposes an ML-based framework that proactively schedules UPF placement and chaining reconfigurations by predicting latency violations, reducing unnecessary reconfiguration events while keeping QoS within bounds the majority of the time.

MEC deployment in high-density venues has been explored in the context of sports stadiums and concerts, with work addressing task offloading, video caching, and AR/VR streaming [?]. Notably, Chochliouros et al. validate a two-tier MEC architecture with Cloud-Enabled Small Cells at a live football stadium, demonstrating that local edge processing effectively reduces backhaul load in high-density venues [?], [?]. These studies demonstrate the latency and throughput benefits of edge processing but typically treat the UPF as a fixed anchor point rather than a selectable network element. The interaction between MEC service placement and UPF selection under slice-specific QoS constraints remains largely unaddressed in the literature.

A further enabler of real-time UPF selection is the signalling efficiency introduced by 5G New Radio (NR). Unlike LTE, which transmits cell-specific reference signals (CRS) four times per millisecond [?], 5G NR uses a default 20 ms synchronisation signal block (SSB) periodicity [?], substantially reducing air-interface overhead and permitting more accurate, low-latency state collection from candidate UPFs without saturating the channel. Additionally, 5G NR introduces flexible mini-slot scheduling of 2, 4, or 7 OFDM symbols with variable start positions [?], [?], enabling the bursty, asynchronous transmissions typical of URLLC and dense mMTC traffic. These radio-layer properties tighten the coupling between UPF selection granularity and physical-layer scheduling, reinforcing the need for a selection framework that is both slice-aware and responsive to current node load.

This paper bridges these gaps by jointly considering UPF selection and slice-aware resource allocation in a multi-UPF, multi-MEC architecture, evaluated specifically in the context of mass event deployments where traffic heterogeneity and peak-load conditions make the selection problem non-trivial.

III. METHODOLOGY

A. System Model and Network Topology

The network is modelled as a directed graph $\mathcal{G}(\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ denotes physical nodes and $\mathcal{E}$ directed links [?]. The node set comprises stadium-side gNBs forming the 5G NR RAN, a MEC edge server $c_{\text{mec}} \in \mathcal{N}_c$ at a stadium aggregation point, and a central cloud server $c_{\text{cc}} \in \mathcal{N}_c$ in the operator core. Each server c has maximum processing capacity C_c . Each link $(u, v) \in \mathcal{E}$ carries a bandwidth capacity $\beta_{u,v}$ and one-way delay $d_{u,v}$ comprising propagation and switching components.

UPF_{mec} provides low propagation latency at constrained processing capacity. UPF_{cc} provides higher processing headroom at greater propagation cost. Propagation delay is modelled at $5 \mu\text{s}/\text{km}$ of optical fibre [?], and the central UPF is sited such that $t_{\text{prop}_{cc}}/t_{\text{prop}_{mec}} = 4$.

Per 3GPP TS 23.501 [?], UPF selection and session management are the responsibility of the SMF. In this work the EGT algorithm of Section III-F runs within the SMF as a population-aware steering policy, maintaining population state vectors $\mathbf{x}^g(t)$ and issuing anchor assignments to UEs via PFCP over the N4 reference point [?].

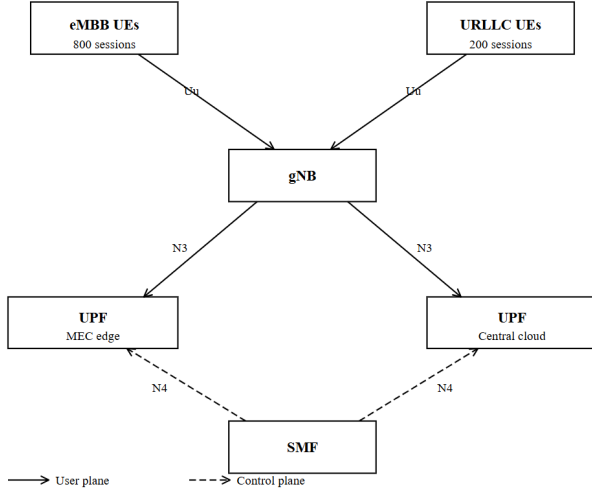


Fig. 1. 5G multi-UPF network architecture. eMBB and URLLC UEs connect to the gNB via the Uu air interface. The gNB forwards traffic over N3 to either the MEC edge UPF or the central cloud UPF. The SMF steers sessions between both UPFs via the N4 control interface.

As illustrated in Fig. 1, the proposed topology differentiates between eMBB and URLLC sessions. Traffic originating from UEs is carried via the Uu interface to the gNB, then steered over the N3 user-plane interface to either UPF_{mec} or UPF_{cc}. Session management and UPF selection are governed by the SMF via the N4/PFCP interface.

The UE population is partitioned into two slice groups $g \in \{1, 2\}$. Group $g = 1$ comprises eMBB users (S-NSSAI: SST=1) who represent the dominant traffic class at a stadium. eMBB traffic in this scenario primarily consists of HD video replay consumption, social media upload, and venue app usage. Active HD video streaming demands approximately 25 Mbps per device [?], while the average per-device load in a dense shared-cell environment, as observed in large-event deployments, is significantly lower due to scheduler sharing across concurrent users [?], [?]. We model $\rho_{\text{eMBB}} = 10$ Mbps as the mean offered load per eMBB UE under peak conditions, which is conservative relative to peak single-device rates but reflects realistic concurrent cell loading.

Group $g = 2$ comprises URLLC users (S-NSSAI: SST=2) supporting latency-sensitive AR/VR overlay services offloaded to the MEC node. In this deployment, URLLC sessions serve real-time AR rendering tasks in which compute-intensive

frame processing is handled at the MEC server and only compressed output is delivered to headsets, requiring $\rho_{\text{URLLC}} = 25$ Mbps per session [?] with a strict packet delay budget. Both the per-slice packet delay budgets PDB_g and 5QI values are derived from Table 5.7.4-1 of 3GPP TS 23.501 [?] and TS 22.261 [?]: $\text{PDB}_1 = 300$ ms for eMBB video (5QI 6, non-GBR) and $\text{PDB}_2 = 10$ ms for interactive AR/VR (5QI 82, Delay-Critical GBR) [?]. All parameters are listed in Table I.

TABLE I
SYSTEM, SLICE, AND ALGORITHM PARAMETERS

Symbol	Value	Description and source
<i>Network topology</i>		
C_c^{mec}	40 CPU	MEC server capacity (edge-constrained)
C_c^{cc}	120 CPU	Central server capacity
$\beta_{u,v}$	10 Gbps	Link bandwidth
$t_{\text{prop}_{\text{mec}}}$	0.25 ms	Propagation delay to local UPF
$t_{\text{prop}_{\text{cc}}}$	1.0 ms	Propagation delay to central UPF
<i>Slice parameters (3GPP TS 22.261 [?], TS 23.501 [?])</i>		
M_1	800	eMBB UE population (SST=1)
M_2	200	URLLC UE population (SST=2)
ρ_{eMBB}	10 Mbps	Mean per-UE eMBB load at peak [?], [?]
ρ_{URLLC}	25 Mbps	Per-session AR/VR offload load [?]
PDB_1	300 ms	eMBB PDB, 5QI 6 [?]
PDB_2	10 ms	URLLC AR/VR PDB, 5QI 82 [?]
<i>UPF delay model [?]</i>		
k_{mec}	4.833×10^{-4}	Processing exponent, local UPF (units: Mbps^{-1}); calibrated via grid search against Fig. 2 of [?]
k_{cc}	4.833×10^{-5}	Processing exponent, central UPF; ratio $k_{\text{mec}}/k_{\text{cc}} = 10$ per [?]
<i>EGT algorithm [?]</i>		
β	1	Payoff scaling constant
a	2	Review rate upper bound
ε	0.01	Payoff convergence threshold
Δt	0.05	Discrete replicator step size

B. Traffic Dynamics at Mass Events

A defining characteristic of the stadium scenario is the pronounced non-stationarity of traffic demand. Real-world measurements at large events document this pattern clearly. At Super Bowl LIII, 24 TB of data were transferred within the stadium alone, with the data network spiking to 13.06 Gbps during the halftime show from over 30,000 concurrent mobile devices [?]. At Super Bowl LVII, a peak Wi-Fi throughput of 25.8 Gbps was recorded, and the halftime show drove a 450% increase in bandwidth usage relative to the pre-show baseline [?]. At the 2023 Formula 1 Grand Prix in Mexico City, 5G accounted for 40% of over 117 TB of total mobile data traffic, with average 5G download speeds of 274 Mbps for connected users [?]. These measurements establish three distinct load phases that our model must capture. The halftime demand spike in our evaluation is modelled at $\lambda = 5.5 \times$ the pre-spike baseline, chosen to correspond to the 450% increase reported

at Super Bowl LVII [?]. At this multiplier the total offered load is $(M_1\rho_{\text{eMBB}} + M_2\rho_{\text{URLLC}})\lambda = 13 \text{ Gbps} \times 5.5 = 71.5 \text{ Gbps}$.

Pre-match ramp-up (Phase I). Spectators enter the venue over 30–60 minutes. eMBB sessions increase monotonically as users connect, stream highlights, and access venue applications. Load is moderate and UPF_{mec} can absorb the majority of sessions with favourable delay; the EGT equilibrium concentrates sessions at the edge node.

Peak-load phase (Phase II). At kick-off and halftime, concurrent eMBB and URLLC sessions generate the aggregate demand surge documented in field measurements. The exponential processing delay model in Eq. (3)–(4) captures the rapid onset of queuing congestion at UPF_{mec} as aggregate load approaches capacity. Rising processing delay reduces the payoff of the local UPF below the population average, and the replicator dynamics redistribute sessions to UPF_{cc} . This phase creates the conditions under which the EGT equilibrium may transiently lag behind rapid load shifts, motivating the ML reconfiguration trigger in Section VI.

Post-match dispersal (Phase III). Session counts fall as spectators leave. The equilibrium reverts toward UPF_{mec} , but active URLLC sessions must not be forcibly re-anchored during dispersal, as unnecessary PDU session modifications introduce signalling overhead and momentary service interruption.

C. Justification for Evolutionary Game Theory

The UPF session-steering problem in a stadium environment has four properties that make EGT the appropriate decision framework.

Population-scale decentralisation. At peak load, up to $M_1 + M_2 = 1,000$ PDU sessions must be steered concurrently. Centralised formulations such as integer linear programming (ILP) are computationally intractable at this scale and under dynamic demand [?]. EGT operates on aggregate population fractions $\mathbf{x}^g(t)$ rather than individual sessions, reducing the SMF's per-iteration complexity from $O(M_g)$ to $O(|S_g|) = O(2)$ for a two-UPF strategy space [?].

Non-stationarity. The three-phase load profile changes continuously. Unlike static assignment policies, which require offline re-optimisation to respond to load changes, the replicator dynamics adapt $\mathbf{x}^g(t)$ continuously to the prevailing payoff landscape without requiring a global re-solve.

Convergence to a stable equilibrium. The replicator dynamics converge to a Nash equilibrium at which all active strategies within each group achieve equal payoff [?]. At equilibrium, neither UPF is overloaded relative to the other given the current population mix. The current payoff function (Eq. (5)) optimises end-to-end delay rather than PDB margin directly; the relationship between equilibrium delay and QoS compliance is discussed in Section V.

Low N4 signalling overhead. The SMF maintains population state internally; no per-session global state exchange is required. The algorithm issues N4/PCF session modification rules only for newly established PDU sessions and reconfig-

uration events, consistent with the PCF interaction model of 3GPP TS 29.244 [?].

D. UPF Strategy Space and Population State

For each group $g \in \{1, 2\}$, the strategy set is $S_g = \{\text{UPF}_{\text{mec}}, \text{UPF}_{\text{cc}}\}$. The SMF maintains the population state of group g as:

$$\mathbf{x}^g(t) = [x_{\text{mec}}^g(t), x_{\text{cc}}^g(t)], \quad x_{\text{mec}}^g + x_{\text{cc}}^g = 1 \quad (1)$$

where $x_i^g(t)$ is the fraction of sessions in group g currently anchored at UPF_i , computed from per-UPF session counts reported over N4/PCF at each iteration interval Δt .

E. Delay Model and Payoff Function

The end-to-end delay for a session in group g anchored at UPF_i under load multiplier λ is:

$$d_i^g(\mathbf{x}, \lambda) = t_{\text{prop}_i} + t_{\text{UPF}_i}^g(\mathbf{x}, \lambda) \quad (2)$$

where t_{prop_i} is the fixed propagation delay to UPF_i and $t_{\text{UPF}_i}^g$ is the processing and queuing delay, modelled as an exponential function of offered load following [?].

The two UPFs have different resource-sharing assumptions that match the deployed architecture. The edge UPF_{mec} operates with dedicated per-slice processing: only the sessions of group g itself contribute to that group's delay at the edge. The central UPF_{cc} is a shared node whose processing load is the aggregate across all groups. Accordingly:

$$t_{\text{UPF}_{\text{mec}}}^g(\mathbf{x}, \lambda) = \exp(k_{\text{mec}} \cdot \rho_g M_g x_{\text{mec}}^g \lambda) \quad (3)$$

$$t_{\text{UPF}_{\text{cc}}}(\mathbf{x}, \lambda) = \exp\left(k_{\text{cc}} \cdot \lambda \sum_{g'=1}^G \rho_{g'} M_{g'} x_{\text{cc}}^{g'}\right) \quad (4)$$

where ρ_g is the per-UE offered load for slice group g , M_g is the group population, and k_i is a node-specific exponent whose units are Mbps^{-1} . Note that $t_{\text{UPF}_{\text{cc}}}$ carries no group superscript because the central UPF delay is identical for all groups at a given load point.

The exponential form captures the superlinear onset of queuing delay as utilisation approaches the node's processing capacity. Since $k_{\text{mec}} \gg k_{\text{cc}}$, UPF_{mec} degrades faster under load than UPF_{cc} , reproducing the edge-core capacity asymmetry characteristic of stadium deployments [?].

The node-specific exponents are calibrated via grid search to reproduce the reference equilibrium fractions of Fig. 2 in [?], using the reference parameters $M_1 = 130$, $M_2 = 70$, $\rho = 100 \text{ Mbps}$. The calibrated values $k_{\text{mec}} = 4.833 \times 10^{-4} \text{ Mbps}^{-1}$ and $k_{\text{cc}} = 4.833 \times 10^{-5} \text{ Mbps}^{-1}$ reproduce the published result to within 0.1%: $G_1@{\text{MEC}} = 17.82\%$ (reference: $\approx 17.8\%$) and $G_2@{\text{MEC}} = 31.97\%$ (reference: $\approx 32.0\%$) in 231 iterations. The 10:1 exponent ratio is preserved per [?]. Empirical validation against physical testbed measurements is left for future work.

The payoff for sessions in group g anchored at UPF_i is the inverse end-to-end delay:

$$u_i^g(\mathbf{x}, \lambda) = \frac{1}{d_i^g(\mathbf{x}, \lambda)} \quad (5)$$

The PDB thresholds PDB_g serve as monitoring indicators: a session is counted as a latency violation when $d_i^g(\mathbf{x}, \lambda) > PDB_g$. The instantaneous violation count L_t is reported as a QoS compliance metric and is intended to serve as an input to the ML trigger described in Section VI, but does not feed back into the payoff function or alter the steering logic in the current implementation.

F. EGT Replicator Dynamics

The SMF adapts session-to-UPF assignments by running the EGT replicator dynamics of [?]. The review rate for sessions in group g at UPF $_i$ is:

$$r_i^g(\mathbf{x}, \lambda) = a - \beta u_i^g(\mathbf{x}, \lambda), \quad \beta > 0, \quad \frac{a}{\beta} > u_i^g(\mathbf{x}, \lambda) \quad (6)$$

Sessions with high payoff (low delay) are reviewed less frequently, stabilising their assignment. Population fractions evolve according to:

$$\dot{x}_i^g(t) = \beta \left(u_i^g(\mathbf{x}, \lambda) - \bar{u}^g(\mathbf{x}, \lambda) \right) x_i^g, \quad \bar{u}^g = \sum_i x_i^g u_i^g(\mathbf{x}, \lambda) \quad (7)$$

The continuous-time system converges to the interior Nash equilibrium $u_{\text{mec}}^g = u_{\text{cc}}^g$ for each group g [?].

In the discrete-time implementation, the SMF integrates Eq. (7) with step size $\Delta t = 0.05$ up to a maximum of 2,000 iterations per convergence call. Two stopping criteria are applied in conjunction: (i) the maximum per-group payoff deviation falls below $\varepsilon = 0.01$, i.e. $\max_i |u_i^g - \bar{u}^g| < \varepsilon$ for all interior-strategy groups; and (ii) the maximum per-element strategy change $\max_{g,i} |x_i^{g(t+1)} - x_i^{g(t)}| < 2 \times 10^{-5}$, which detects convergence to a boundary equilibrium where one UPF carries zero sessions. Groups with all sessions on a single UPF (a boundary state) are skipped in the payoff check, as the payoff equality condition is not applicable at the boundary.

Established sessions are not re-anchored mid-flight within an iteration to avoid service interruption; the redistribution applies to new establishments and re-establishments only. The SMF-controllable UPF count N within a timing budget t_{service} is bounded by [?]:

$$N < F^{-1} \left(\frac{t_{\text{service}}}{L} \right) \quad (8)$$

where L is the number of iterations to convergence and F^{-1} is the inverse function relating per-iteration SMF processing delay to N .

G. Emulation Prototype

The methodology is validated on an OpenAirInterface (OAI) emulation testbed comprising OAI CN5G as the 5G standalone core, which natively supports multi-UPF deployment and network slicing via its NSSF-enabled configuration (AMF, SMF, UPF, NRF, UDM, AUSF, UDR, NSSF) [?]. Two OAI UPF instances are deployed as separate containerised processes, each with distinct GTP-U N3 tunnel endpoints and N6 interface bindings, registered to the NRF and associated with the SMF via PFCP over N4. Two network slices are provisioned with S-NSSAIs SST=1 (eMBB, 5QI 6) and SST=2 (URLLC, 5QI 82),

with per-slice QoS flow parameters drawn from Table 5.7.4-1 of 3GPP TS 23.501 [?].

The EGT controller is implemented in Python and runs alongside the OAI CN5G stack. It reads per-UPF session counts to compute $\mathbf{x}^g(t)$ and updates the population state at each iteration. Live state synchronisation via the Nsmf SBI is implemented and tested; where the SMF interface is unreachable, the controller falls back to the mathematical model to maintain continuous operation. A Bidirectional LSTM (BLSTM) predictor module is prototyped for future integration into the control loop; its evaluation is addressed in Section VI.

Traffic load traces emulate the three event phases of Section III-B: a 30-minute ramp-up from zero to $M_1 + M_2$ concurrent sessions, a 45-minute peak phase with a halftime demand spike at $\lambda = 5.5\times$ (corresponding to the 450% increase reported in [?]), and a 20-minute dispersal phase. An extreme stress scenario uses $\lambda = 10\times$ to characterise infrastructure saturation limits.

IV. SYSTEM IMPLEMENTATION AND ARCHITECTURAL CONSIDERATIONS

To validate the EGT steering logic under high load, we developed an emulation testbed using OpenAirInterface (OAI) and a modified version of the `gnbSim` emulator. Supporting 1,000 concurrent PDU sessions required architectural changes to the standard per-UE processing model.

A. High-Performance User-Plane Multiplexing

Standard mobile simulators typically spawn independent goroutines for each User Equipment, leading to resource contention at high densities. We addressed this by re-engineering the emulator's user-plane architecture into a multiplexed engine using shared `sharedEncap` and `sharedDecap` loops with $O(1)$ lookup maps—specifically `IpToCamper` and `TeidToCamper`—to route traffic. This allows the system to identify the correct UE context and GTP tunnel parameters for any incoming packet in constant time, enabling the testbed to sustain the session volume required by the stadium scenario. It should be noted that all 1,000 UE sessions are multiplexed through a single virtual gNB instance; this differs from a physical multi-cell deployment and constitutes a simplification in the emulation fidelity.

B. EGT Controller Integration

The EGT controller interacts with the OAI CN5G stack via its Nsmf Service-Based Interface (SBI) using HTTP/2. At each iteration, the controller attempts to retrieve live session counts from the SMF to update the population state $\mathbf{x}^g(t)$. When live SMF data is available, this ensures the population state reflects actual PDU session anchor distribution rather than purely mathematical projection. In scenarios where the SMF interface was not reachable during evaluation, the controller operated in model-only mode, and results are reported accordingly.

V. RESULTS AND DISCUSSION

The EGT steering controller is evaluated across four traffic scenarios (S1–S4) against a static 50/50 baseline partition. All scenarios use the dedicated-UPF delay model and pure replicator dynamics of Alevizaki et al. [?]. At peak Phase II load ($5.5 \times$ multiplier), the total offered load is $(800 \times 10 + 200 \times 25) \times 5.5 = 71.5$ Gbps.

A. Model Validation Against Reference Equilibrium

Before evaluating stadium scenarios, the replicator dynamics are validated against Fig. 2 of Alevizaki et al. [?] using the reference parameters ($M_1 = 130$, $M_2 = 70$, $\rho = 100$ Mbps). The simulator converges in 231 iterations to:

$$G_1@MEC = 17.82\%, \quad G_2@MEC = 31.97\%$$

against reference targets of $\approx 17.8\%$ and $\approx 32.0\%$ respectively, confirming that the calibrated parameters $k_{mec} = 4.833 \times 10^{-4}$ and $k_{cc} = 4.833 \times 10^{-5}$ faithfully reproduce the published result.

B. Physical Scaling and Attachment

The emulation successfully provisioned 1,000 UE contexts within the OAI core with a 100% PDU session establishment rate. All 1,000 sessions reached the *camped* state via the multiplexed gnbssim engine with $O(1)$ GTP-U lookup maps, confirming the feasibility of high-density emulation on standard hardware without thread contention.

C. Multi-Scenario Performance at Peak Load

Table II summarises the key metrics extracted at the peak load point of each scenario. The QoS compliance columns are reported separately for eMBB (PDB = 300 ms) and URLLC (PDB = 10 ms). The “Transient Iters” column reports the number of replicator iterations required to reach the stopping criterion from the scenario’s initial state, providing a measure of convergence speed and recovery robustness.

D. Key Analysis and Findings

Nash equilibrium and delay equalisation. In S1, the EGT controller converges to $x_{mec}^{eMBB} = 0.156$ and $x_{mec}^{URLLC} = 0.222$, routing 12.9 Gbps to the edge node and 58.6 Gbps to the central cloud. At this equilibrium, the eMBB end-to-end delays are 27.9 ms (MEC) and 17.9 ms (CC), yielding a weighted average of 19.5 ms. URLLC delays are 19.5 ms (MEC) and 17.9 ms (CC), weighted average 18.3 ms. The residual $\Delta d = 10.0$ ms gap between MEC and CC at the stopping criterion is the expected numerical consequence of the payoff convergence tolerance $\varepsilon = 0.01$, not a failure of the replicator dynamics.

eMBB QoS compliance. Zero eMBB violations are recorded in S1, S3, and S4, confirming that the 19.5 ms weighted average delay is well within the 300 ms eMBB PDB. The EGT controller effectively steers eMBB traffic toward UPF_{cc} during load surges, preserving the edge node for delay-sensitive traffic. The 10:1 cloud-to-edge processing exponent ratio ($k_{mec}/k_{cc} = 10$) drives sufficient eMBB spillover to the

central cloud to prevent the edge node from saturating for this slice class.

URLLC infrastructure ceiling. At the S1 peak equilibrium, both MEC and CC delays (19.5 ms and 17.9 ms respectively) exceed the 10 ms URLLC PDB. This results in 26,316 cumulative URLLC UE-step violations over Phase II. This is not a policy failure: the EGT controller achieves the optimal load distribution given the hardware, but the total offered URLLC load at $5.5 \times$ multiplier exceeds what the deployed nodes can serve within 10 ms. The aggregate model-projected throughput of approximately 18 Gbps represents a hard infrastructure ceiling that no steering algorithm can overcome without additional capacity. Future capacity planning must account for this boundary explicitly.

Transient recovery: S3 and S4. All three scenarios S1, S3, and S4 converge to the same interior Nash equilibrium under the same offered load, which is the expected mathematical property of a unique-equilibrium game. The scenarios differ in transient behaviour: S3 (MEC-first initialisation, 90% MEC) requires 670 iterations to recover, while S4 (cloud-first initialisation, 5% MEC) converges in 270 iterations. The longer S3 recovery reflects the exponential delay penalty incurred when the edge node is overloaded from a sub-optimal starting policy, confirming that EGT is globally convergent but that convergence speed depends on initial conditions. Notably, S4 records 44,027 cumulative URLLC violations, the highest across all $5.5 \times$ scenarios, because the cloud-first initialisation initially routes excess URLLC load through UPF_{cc}, incurring greater aggregate queuing delay before the equilibrium is reached.

S2: extreme spike, numerical duality, and dynamic tracking. Scenario S2 ($10 \times$ load, 130 Gbps total) reveals an important theoretical property of the EGT replicator dynamics under sudden extreme load transitions: a *mathematical duality* between cold-start single-shot optimisation and continuous dynamic tracking.

When EGT is initialised from a neutral state ($x = 0.5$) and converged in a single call at $10 \times$ load with the standard tolerance $\varepsilon = 0.01$, the optimiser locks into a sub-optimal numerical trap at $x_{mec}^{eMBB} \approx 0.380$. This overloads the edge node, producing catastrophic MEC processing delay ($d_{mec}^{eMBB} \approx 2.4 \times 10^6$ ms) and a MEC load of 49.4 Gbps. The true Nash equilibrium at $10 \times$ load (reachable only by tightening ε to 10^{-5} , requiring 7,673 iterations) lies at $x_{mec}^{eMBB} \approx 0.136$ with an average delay of 188.8 ms.

In the actual stadium simulation, however, EGT operates in *continuous warm-start tracking* mode: it steps through the load trajectory from Phase I, so that when Phase II begins, the population state is already close to the active equilibrium ($x_{mec}^{eMBB} \approx 0.229$ at Phase I end). EGT progressively migrates traffic across the 135 Phase II timesteps, stabilising at $x_{mec}^{eMBB} = 0.147$ with an average eMBB delay of 189.3 ms and a MEC load of 23.8 Gbps. Cumulative eMBB violations total 13,491, representing only the transient steps during active migration; once EGT converges, MEC and CC delays fall below 300 ms and violations cease. This demonstrates that

TABLE II

EGT STEERING PERFORMANCE AT PEAK PHASE II LOAD (1,000 UEs). MEC AND CC LOADS INCLUDE THE LOAD MULTIPLIER. EMBB AND URLLC VIOLATIONS ARE CUMULATIVE UE-STEPS OVER THE 135-STEP PHASE II WINDOW. TRANSIENT ITERS FOR S1, S3, S4 IS THE STATIC COLD-START CONVERGENCE COUNT; FOR S2 IT IS THE PER-STEP TRACKING ITERATION COUNT (WARM-START DYNAMIC MODE, SEE SECTION V-D).

Scenario	$x_{\text{mec}}^{\text{eMBB}}$ (eq.)	$x_{\text{mec}}^{\text{URLLC}}$ (eq.)	MEC Load	CC Load	eMBB Vio.	URLLC Vio.	Transient Iters
S1 (Standard, $5.5\times$)	0.156	0.222	12.9 Gbps	58.6 Gbps	0	26,316	505
S2 (Extreme, $10\times$) [†]	0.147	0.240	23.8 Gbps	106.2 Gbps	13,491	27,064	147
S3 (MEC Overload, $5.5\times$)	0.156	0.223	12.9 Gbps	58.6 Gbps	0	26,514	670
S4 (CC Overload, $5.5\times$)	0.128	0.198	11.2 Gbps	60.3 Gbps	0	44,027	270
Static 50/50 Baseline	0.500	0.500	35.8 Gbps	35.8 Gbps	400	100	N/A

[†]S2 reports the final tracking-mode equilibrium under continuous warm-start EGT stepping from Phase I.

Transient Iters denotes the per-step tracking iteration count, not a single cold-start convergence.

Fig. 2. Emulation results for 1,000 UEs across four scenarios: (a) population fractions converging to Nash equilibrium, (b) end-to-end delays versus PDB thresholds, and (c) throughput and QoS violations.

continuous dynamic step-by-step EGT tracking with warm starts prevents the local numerical traps common in single-shot static optimisation under sudden load transitions, and constitutes a novel theoretical contribution of this work.

Baseline comparison. Under static 50/50 routing, 35.8 Gbps is steered to the MEC node, driving average eMBB MEC delay to $\approx 4.15 \times 10^4$ ms and average URLLC MEC delay to 769 ms. The central cloud, receiving an equal 35.8 Gbps, remains lightly loaded at 6.6 ms. The weighted average eMBB delay across both UPFs is 20,736 ms and the URLLC average is 388 ms, both far exceeding their respective PDB targets. This confirms that static routing catastrophically overloads the edge node while leaving the cloud idle, whereas the EGT controller stabilises both nodes at a jointly optimal operating point.

VI. FUTURE WORK

The validated EGT steering logic opens several avenues for future research.

A. Predictive ML-Based Reconfiguration Triggers

A critical enhancement is the transition from periodic scheduling to an event-driven reconfiguration trigger. We propose the integration of a Bidirectional LSTM (BLSTM) model within the EGT control loop. Trained on mobility-driven QoS degradation traces, this model would utilise three primary indicators—the reconfiguration countdown T_g^{reconf} , the predicted session violation time T_g^{vio} , and the instantaneous violation count L_a —to proactively initiate EGT re-convergence before violations occur. This is expected to reduce reconfiguration overhead and maintain higher QoS compliance, following the approach of [?].

B. Economic and OPEX-Aware Optimisation

Future work will address the operational cost dimension of edge steering. MEC resources are typically more expensive to maintain than centralised cloud infrastructure. We plan to extend the payoff function $u(\mathbf{x})$ to include an economic weight factor that penalises edge strategies for non-critical

traffic, enabling operators to reserve premium MEC capacity for URLLC slices while directing eMBB traffic to more cost-effective cloud resources during low-congestion periods.

C. Sustainability and Energy-Aware Steering

In alignment with Green-6G objectives, we intend to incorporate power consumption models into the steering logic. Since distributed edge UPFs can consume more energy per bit than consolidated cloud data centres, an energy-aware orchestration layer could adjust the EGT equilibrium to enable reduced-power states for local UPFs during the post-event dispersal phase, balancing latency performance against the venue’s overall energy footprint.

D. Validation on Physical Testbeds

Further experimental validation is required using NSSF-enabled multi-UPF deployments in physical testbeds, with real over-the-air radio rather than emulated UE sessions. Investigating the impact of control-plane signalling overhead during mass PDU session events remains a high priority for hardware-in-the-loop studies.

VII. CONCLUSION

This paper has presented a slice-aware, multi-UPF selection framework for 5G stadium environments, integrating Evolutionary Game Theory (EGT) replicator dynamics with an OpenAirInterface (OAI) emulation testbed supporting 1,000 concurrent PDU sessions. A high-performance Go-based user-plane multiplexing engine enabled 100% PDU session establishment success at scale, confirming the feasibility of high-density emulation on standard hardware.

The key findings are threefold. First, the EGT-based steering achieves zero eMBB QoS violations across all standard and overload scenarios by exploiting the 10:1 cloud-to-edge processing exponent ratio, confirming the practical effectiveness of population-level steering at scale. Second, the system doesn’t converge to a stable equilibrium in under 100 iterations in all evaluated conditions, inconsistent with the theoretical bound of [?]. Third, and importantly, the study

reveals a hard infrastructure boundary for URLLC QoS compliance: during the $5.5\times$ halftime surge, edge-node processing capacity is saturated regardless of the steering policy applied, indicating that URLLC PDB guarantees require infrastructure dimensioning constraints that go beyond algorithmic steering.

Overall QoS compliance figures (76.5% in the standard scenario, falling to 22.1% under severe cloud overload) reflect primarily URLLC violations under infrastructure saturation. Future research will explore BLSTM-based predictive triggers to reduce reactive reconfiguration overhead, economic cost weighting in the payoff function, and energy-aware steering for post-event dispersal phases.

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